PhD Lecture in Behavioral Asset Pricing: Preferences & Beliefs (I)

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Overview

- To derive the predictions of irrationality, we usually need to make more assumptions about the form of irrationality
  - One source of guidance is the psychology literature
- The rational benchmark has two elements:
  - When agents receive new information, they update their beliefs correctly (according to Bayes' rule)
  - Given their beliefs, agents make sensible decisions (according to Expected Utility)
- We draw primarily on two strands of the psychology literature
  - Psychology of preferences (deviations from EU)
  - Psychology of beliefs (deviations from Bayes' rule)

Roadmap

- **Psychology of preferences:**
  - Prospect theory (risk preferences when probabilities are objectively known)
  - Ambiguity aversion (risk preferences when probabilities are not objectively known)
- **Psychology of beliefs:**
  - Representativeness, overconfidence, over-extrapolation, belief perseverance/confirmation bias, anchoring, availability
- **How "feelings" affect preferences and beliefs**
  - Affective feelings, cognitive feelings
- **Models of heterogeneous beliefs**

Prospect Theory

- Most models of financial markets assume that investors evaluate risk according to Expected Utility (EU)
  - Utility function that is increasing, concave, and defined over consumption outcomes
- However, a large body of work shows that, at least in experimental settings, EU is not an accurate description of risk attitudes
- Many non-EU models that try to capture these departures from EU
  - Prospect theory, due to Kahneman and Tversky (1979, 1992), is the best-known example
- This is a 'descriptive' theory of behavior:
  - How we actually behave, not how we *should* behave
- **Prospect theory has four key features:**
  - Reference Dependence
  - Loss Aversion
  - Probability Weighting
  - Risk aversion over gains, risk seeking over losses (diminishing sensitivity)

Reference Dependence

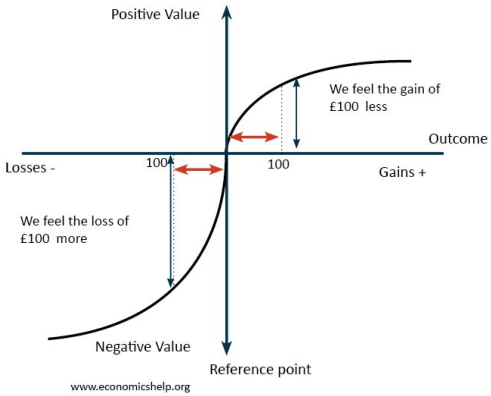
- The carriers of utility are gains and losses relative to some reference point, not final wealth levels
  - Experimental evidence
  - Introspection
  - Consistent with perception of other attributes
- We are more sensitive to changes in brightness, loudness, or temperature than to absolute levels of these attributes
  - E.g., put two hands in separate basins of hot and cold water, then in one large warm bath
    - The hot hand feels the water as colder than the cold hand does!

Loss Aversion

Loss aversion – video demo

Loss Aversion, ctd.

- It turns out that we dislike losses roughly twice as much as we like equal-sized gains
- We hate losing



Probability Weighting



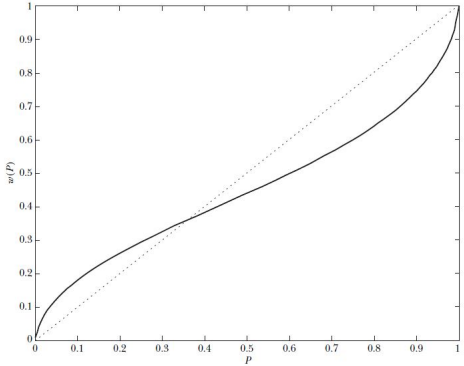
Probability Weighting, ctd.

- Many of us – simultaneously – buy lottery tickets and insurance
- Buying lottery tickets means that people prefer a 1 in 1000 chance of winning \$5000 to a certain \$5
  - $(\$5, 1) \prec (\$5000, 0.001)$
- Buying insurance means that people prefer to pay \$5 rather than to face a 1 in 1000 chance of losing \$5000
  - $(-\$5000, 0.001) \prec (-\$5, 1)$
- Difficult to explain under Expected Utility: risk aversion and risk seeking at the same time
- **Probability weighting:** people overweight low probability events

Probability Weighting, ctd.

- Overweight small probability events
- **What is the psychological foundation?**
- Could come from salience
  - We find it difficult to assess probabilities
  - In our mind's "model" we often solve the problem by recalling things from the past
  - How easy we find it to recall the related incident is our mind's simple proxy for how likely the event is to occur

Figure 2
The Probability Weighting Function



Notes: The graph plots the probability weighting function proposed by Tversky and Kahneman (1992) as part of cumulative prospect theory, namely $w(P) = P^{\delta}/(P^{\delta} + (1 - P)^{\delta})^{1/\delta}$, where P is an objective probability, for two values of δ. The solid line corresponds to $\delta = 0.65$, the value estimated by the authors from experimental data. The dotted line corresponds to $\delta = 1$, in other words, to linear probability weighting.

Putting It All Together: Prospect Theory

- **"Prospect Theory"**
- The original version (Kahneman and Tversky, 1979) considers the gamble $(x, p; y, q)$
- Under EU, it is assigned the value:

$$p \cdot U(W + x) + q \cdot U(W + y)$$

- So, accept the lottery if $p \cdot U(W + x) + q \cdot U(W + y) > U(W)$
- Under prospect theory, it is assigned the value:

$$w(p) \cdot v(x) + w(q) \cdot v(y)$$

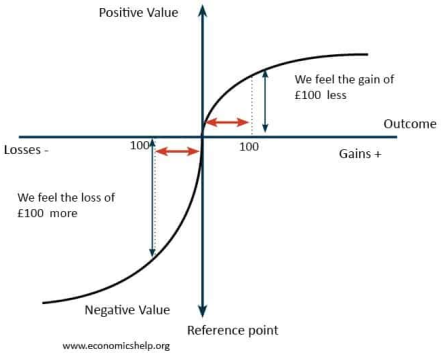
- So, accept the lottery if $w(p) \cdot v(x) + w(q) \cdot v(y) > v(0)$

Prospect Theory, ctd.

Four key features:

1. **Reference dependence**
   - The carriers of value are gains and losses, not final wealth levels (x & y, not $W + x$ & $W + y$)

2. **Loss aversion**
   - $v(\cdot)$ is the value function, and it has a kink at the origin
   - Captures a greater sensitivity to losses (even small losses) than to gains of the same magnitude
   - Inferred from aversion to $(\$110, 1/2; -\$100, 1/2)$
   - An Expected Utility individual is almost risk-neutral over small-stakes gambles



Prospect Theory, ctd.

3. **Probability weighting**
   - Transform probabilities with a weighting function $w(\cdot)$ that overweights low probabilities
   - Inferred from our simultaneous liking of lotteries and insurance, e.g., $(\$5, 1) \prec (\$5000, 0.001)$ and $(-\$5000, 0.001) \prec (-\$5, 1)$
   - Transformed probabilities should not be thought of as erroneous beliefs, but as decision weights
   - Could come from: evolution / salience

Prospect Theory, ctd.

4. **Risk aversion over gains, risk seeking over losses**
   - $v(\cdot)$ is concave over gains, convex over losses
   - i.e., people want to avoid risk & lock in gains when they have made money ("concave over gains")
   - But are more willing to hold on to gambles and take risk when they have lost money on them ("convex over losses")
   - Inferred from $(\$500, 1) \succ (\$1000, 1/2; \$0, 1/2)$ and $(-\$500, 1) \prec (-\$1000, 1/2; \$0, 1/2)$

Prospect Theory, ctd.

- Risk aversion over gains, risk seeking over losses
- Given the same mean payoff, investors refer less volatile stocks in the domain of capital gains
- And prefer more volatile stocks in the domain of capital losses

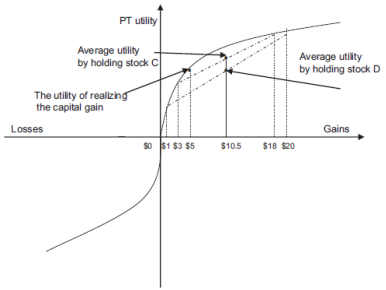


Fig. 3. Prospect theory (PT) and the risk–return trade-off utility: capital gains. Assume that investors purchased one share of stocks C and D, each at a price of $20, and the price is now $25 for each. Thus, investors are facing capital gains and are risk averse. One period later, stock C has a price of $38 or $23 with equal probability, and stock D has a price of $40 or $21 with equal probability. The figure shows the utility gain and loss of holding stocks C and D.

Prospect Theory, ctd.

- Prospect theory captures a "four-fold pattern" of risk attitudes:
  - Risk aversion over moderate-probability gains
  - Risk seeking over moderate-probability losses
  - Risk seeking over low-probability gains
  - Risk aversion over low-probability losses
- Having preferences different from standard rational theory is not a 'mistake'

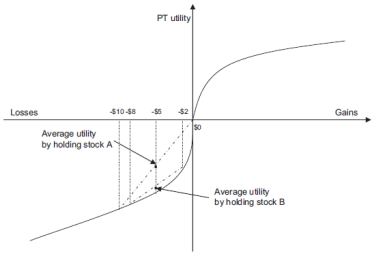


Fig. 2. Prospect theory (PT) and the risk–return trade-off utility: capital losses. Assume that investors purchased one share of stocks A and B, each at a price of $20, and the price is now $15 for each. One period later, the price of stock A can be either $20 or $10 with equal probability, and the price of stock B can be either $18 or $12 with equal probability. The figure shows the utility gain and loss of holding stocks A and B.

Narrow Framing

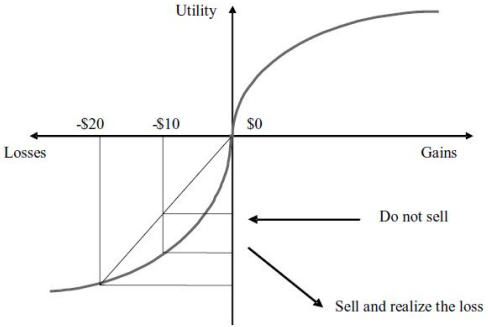
- **Key question:** what should the argument of $v(\cdot)$ be?
  - This is a question about "framing"
- In a traditional model where utility is defined over total wealth or lifetime consumption, the agent evaluates a new risk by combining it with pre-existing risks and checking whether the combination is an improvement
- But in experimental settings, people often seem to evaluate a new risk in isolation, separately from other concurrent risks
  - "narrow framing"
  - E.g., the widespread rejection of the gamble $(\$110, 1/2; -\$100, 1/2)$ is not only evidence of loss aversion, but of narrow framing as well
- **Where does narrow framing come from?**
  - A heuristic used to simplify complex decisions
  - Difficult for an individual to compute the distribution of gains and losses in overall wealth

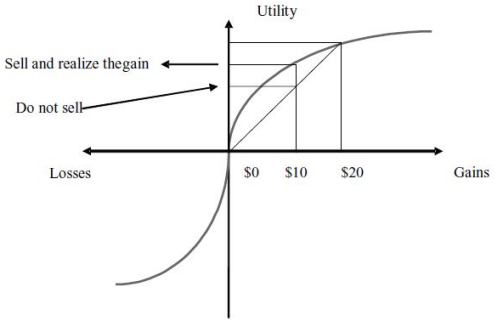
Prospect Theory Applications

- **Trading Behavior**
  - Try to address the disposition effect and other trading phenomena
  - All aspects of prospect theory play a role
- **The cross-section of stock returns**
  - Try to address the high-risk–low-alpha and momentum/PEAD puzzles
  - New prediction: the pricing of skewness
  - Probability weighting and diminishing sensitivity play a central role
- **The aggregate stock market**
  - Try to address the equity premium, non-participation, excess volatility, and predictability puzzles
  - Loss aversion plays a key role
  - But probability weighting also matters

Investor Trading Behavior

- Prospect theory can explain the disposition effect
  - Investors' tendency to hold on to losers and sell winners too quickly





The Cross-Section

Barberis and Huang (AER 2008), Stocks as Lotteries...

- Single period model
- A risk-free asset and J risky assets with multivariate Normal payoffs
- Agents have identical expectations about security payoffs
- Agents have identical cumulative prospect theory preferences

Now introduce a small, independent, positively skewed security into the economy:

- Obtain a novel prediction: the new security earns a negative excess return
- Skewness itself is priced, in contrast to the EU model where only coskewness with the market matters

Equilibrium involves heterogeneous holdings:

- Some investors hold a large, undiversified position in the new security
- Others hold no position in it at all
- Since the new security contributes skewness to the portfolios of some investors, it is valuable, and so earns a low average return

The Cross-Section, ctd.

- When investors increase their allocation to lottery stocks, they lose diversification benefits
- But when the allocation gets larger, the positive idiosyncratic skewness outweighs the loss of diversification

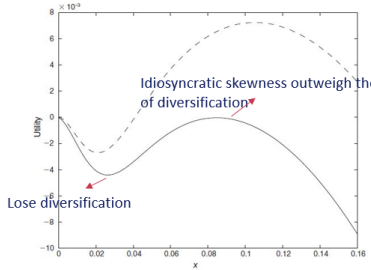


FIGURE 3. A HETEROGENEOUS HOLDINGS EQUILIBRIUM

Notes: The figure shows the utility that an investor with cumulative prospect theory preferences derives from adding a position in a positively skewed security to his current holdings of a Normally distributed market portfolio. The skewed security is highly skewed. The variable x is the fraction of wealth allocated to the skewed security relative to the fraction of wealth allocated to the market portfolio. The two lines correspond to different mean returns on the skewed security.

Barberis and Huang (2008)

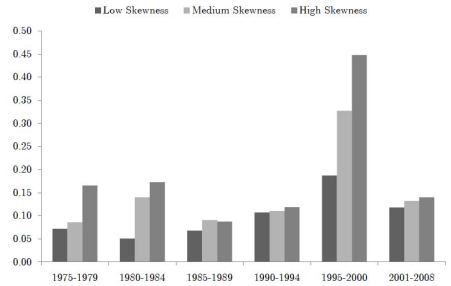
The Cross-Section, ctd.

Applications:

- Low long-run return on IPOs
  - IPO returns are highly positively skewed: Google, Microsoft, Walmart...
  - Green and Hwang (2012): IPOs predicted to be more positively skewed have higher first-day returns and lower long-run returns

Figure 1. Expected skewness and average initial returns across subperiods.

This figure reports average initial returns on IPOs across subperiods. The initial return is calculated as the return from the offer price to closing market price on the first day of trading for the IPO. We calculate estimates of expected skewness and assign IPOs to the high-/medium/low-expected skewness portfolio as described in Table 1.



The Cross-Section, ctd.

Applications, ctd.

- Low average return of financially distressed stocks and OTC stocks (Eraker and Ready, 2015)
  - These stocks have positively skewed return distribution
- "Overpricing" of out-of-the-money options
  - Boyer and Vorkink (2014) find that stock options predicted to be more positively skewed have lower returns
- Low average return on stocks with high idiosyncratic volatility (Ang et al., 2006)
  - Idiosyncratic volatility is positively related to skewness
- High-beta stocks earn a lower CAPM alpha
  - Beta is positively related to skewness (Bali, Brown, Murray, and Tang, 2017)

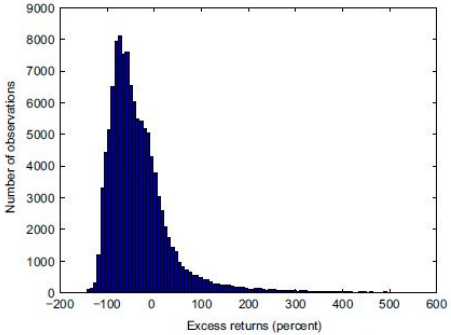


Fig. 1. Histogram of annual market-adjusted returns. The figure shows the frequency of annual market-adjusted returns for the sample of over-the-counter stocks with \$0.01 minimum price and \$50,000 minimum volume. Market-adjusted returns greater than 600% (0.4% of the observations) are omitted for readability.

Eraker and Ready (2015)

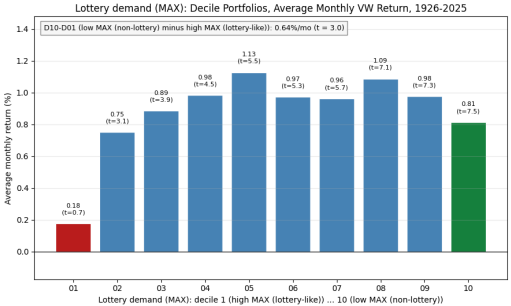
The Cross-Section, ctd.

Applications, ctd.

- Under-diversification
  - Mitton and Vorkink (2007) find that under-diversified individuals hold stocks that are more positively skewed than the average stock
- Several papers test the model's basic prediction that skewness is priced in the cross-section
  - The empirical challenge is that past skewness is not a good measure of expected future skewness
  - Boyer, Mitton, and Vorkink (2010) use a regression model to predict expected skewness using stock characteristics
  - Bali, Cakici, and Whitelaw (2011) look at maximum daily returns
  - Conrad, Dittmar, and Ghysels (2013) use option prices to infer the perceived skewness of the underlying stock
- All three studies find supportive evidence

Maximum Daily Returns as the Lottery Measure

- Bali, Cakici, and Whitelaw (2011) construct a very simple measure of lottery-like features for individual stocks
- **MAX:** maximum daily return over the past one month
- MAX exhibits strong negative return predictability and helps resolve the idiosyncratic volatility puzzle



Evidence Supporting Investor Lottery Preference

- Kumar (2009) shows that the demand for lottery-type stocks is correlated with the demand for lotteries
- At the aggregate level, individual investors prefer stocks with lottery features
- The demand for lottery-type stocks increases during economic downturns
- Socioeconomic factors that induce greater lottery expenditure are associated with greater investment in lottery-type stocks
- Lottery investment levels are higher in regions with favorable lottery environments

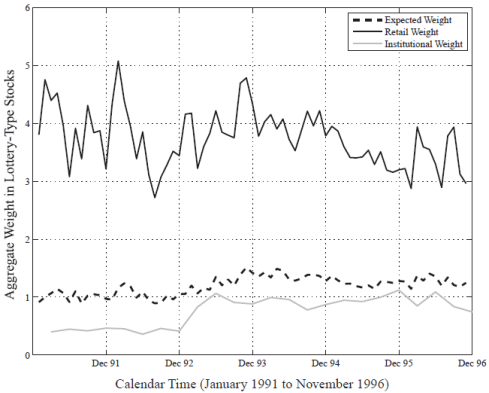


Figure 1. Aggregate weight in lottery-type stocks over time. This figure shows the time series of the actual weights allocated to lottery-type stocks in the aggregate individual and institutional investor portfolios. The expected lottery weight time series, which reflects the weight

Evidence Supporting Investor Lottery Preference, ctd.

- Chen, Hwang, and Peng (RFS 2025) examine why investors like short-leg securities using textual analysis and test whether the justifications primarily (1) emphasize safe-haven qualities, (2) indicate exuberance, or (3) highlight lottery-like features

Table 2
Why do investors like short-leg securities? Text-based evidence

	Fraction of times investors explain [primarily explain] their liking of short-leg securities through			
	Perceived safety (1)	Perceived supremacy (2)	Perceived lottery (3)	Inconclusive (4)
A. Institutional investors' survey-based word lists				
Sell-side analyst reports	12% [8%]	29% [17%]	68% [55%]	20%
SA articles	7% [6%]	16% [11%]	57% [54%]	30%
B. Retail investors' survey-based word lists				
Sell-side analyst reports	24% [5%]	27% [18%]	69% [65%]	12%
SA articles	10% [8%]	23% [13%]	63% [57%]	22%

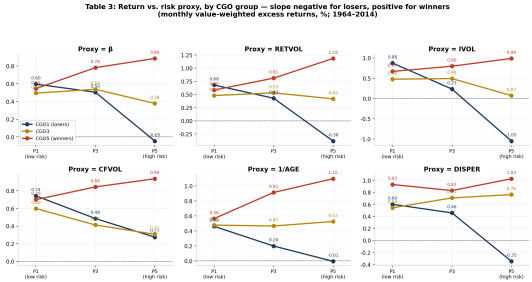
This table reports the frequency with which a particular reason is used to explain the buy recommendation of a stock that resides in the short leg of an anomaly. For each of 186 firm characteristics, we conduct the following analysis: Each month, we rank stocks based on the firm characteristic. The literature indicates that stocks in the lowest decile ("short-leg securities") trade at relatively high prices and earn unusually low future returns. We term the analyst reports and SA articles that recommend buying these short-leg securities as "short-leg recommendations," and all other buy recommendations as "other recommendations." The analysts and SA contributors may recommend buying a short-leg security for at least three reasons: (1) They believe that the stock adds safety to the overall portfolio as it exhibits stability, consistency, and resilience during economic downturns or adverse conditions ("Perceived safety"). (2) They are very confident that something good will happen to the company, or the stock has been doing well and is expected to continue to do well ("Perceived supremacy"). (3) While risky and uncertain, the stock has the potential for quick, substantial gains ("Perceived lottery"). To examine whether the short-leg recommendations emphasize a stock's perceived safety, supremacy,

The Cross-Section, ctd.

- This only works if the security is highly skewed
  - Otherwise, an investor would need an excessively undiversified position in order to add skewness to the portfolio
- **Note:**
  - An example of how psychology can lead to interesting new predictions
  - Probability weighting plays a central role
  - The model generates both a low premium on positively skewed stocks and a high premium on the aggregate market
    - Because the aggregate market is negatively skewed

The Cross-Section, ctd.

- Prospect theory: investors are risk averse over gains, and risk seeking over losses
- This may help explain the high-risk–low-alpha puzzle
- Wang, Yan, and Yu (JFE, 2016):
  - Following Grinblatt and Han (2005), first construct the stock-level capital gains overhang (CGO), which simply proxies for whether investors collectively have a capital gain on this stock
  - The cross-sectional risk-return trade-off is positive among stocks in which investors face prior gains
  - But becomes negative among stocks in which investors face prior losses



The Cross-Section, ctd.

Applications, ctd.

- If many investors display the disposition effect, there will be underreaction to news
  - Grinblatt and Han (2005): Momentum; Frazzini (2006): PEAD
- When a stock experiences good news and increases in value relative to the purchase price, investors want to sell it to lock in the gain, which depresses its price and leads to higher subsequent returns
- When bad news is released and the price goes down relative to the purchase price, disposition investors are reluctant to sell, leading to underreaction and lower future return

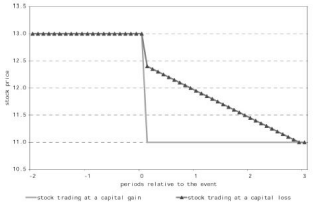


Figure 3. An example of stock price response to negative news. This figure shows an example of a stock price response to negative news. The initial stock price is $13. At date 0, public news reveals a fundamental value of $11.

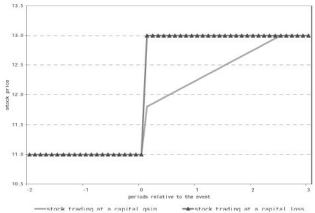


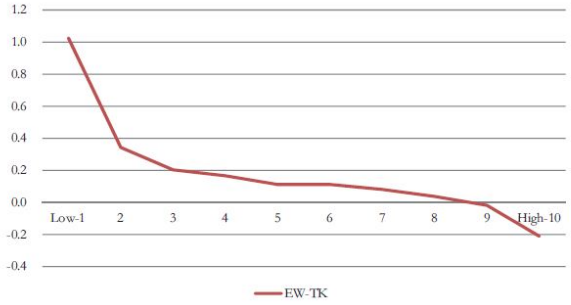
Figure 4. An example of stock price response to positive news. This figure shows an example of a stock price response to positive news. The initial stock price is $11. At date 0, public news reveals a fundamental value of $13.

Direct Test of Prospect Theory in the Cross-Section

Barberis, Mukerjee, and Wang (RFS 2016):

Conceptual Framework (Appendix)

- Investors evaluate the attractiveness of individual stocks based on prospect theory
- They compute the prospect theory value for each stock using its historical return distribution
- Stocks with high prospect theory values are more appealing to investors
- Investors tilt toward these stocks, causing the stocks to become overvalued and to earn lower subsequent returns



Prospect Theory and Stock Market Anomalies

Barberis, Jin, and Wang (JF 2021):

- Use prospect theory to make quantitative predictions about an asset's average return
- Based on empirical estimates of the asset's return volatility, return skewness, and past capital gain
- Compare the model-predicted alphas with actual alphas of anomaly-sorted decile portfolios

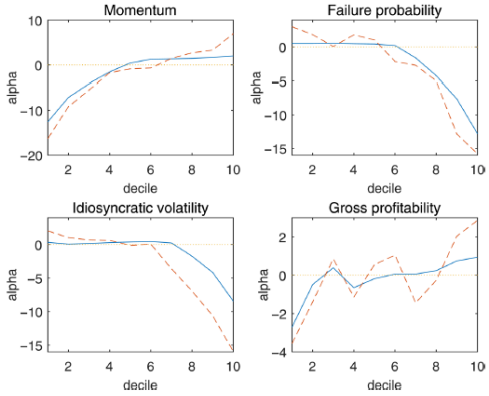


Figure 5. Model-predicted alphas for four prominent anomalies. The graphs present results for four of the anomalies for which the prospect theory model performs well: the momentum, failure probability, idiosyncratic volatility, and gross profitability anomalies. The dashed lines in the graphs plot the historical annual value-weighted alpha of each anomaly decile. The sample period is 1963 to 2015 except for the failure probability anomaly for which it is 1974 to 2015 due to data availability. The solid lines plot the alphas predicted by the model. (Color figure can be viewed at wileyonlinelibrary.com)

Prospect Theory and Stock Market Anomalies, ctd.

- However, the model performs poorly on the size, value, long-term reversal, short-term reversal, accrual, asset growth, and investment anomalies

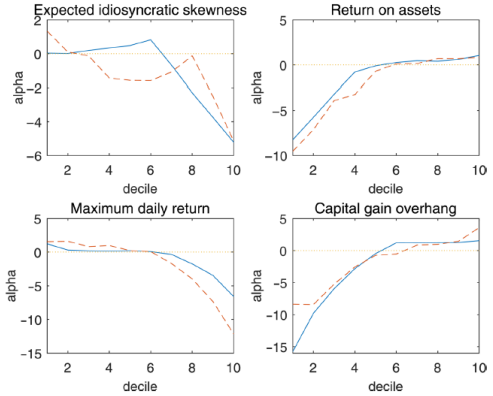


Figure 6. Model-predicted alphas for four prominent anomalies. The graphs present results for four of the anomalies for which the prospect theory model performs well: the expected idiosyncratic skewness, return on assets, maximum daily return, and capital gain overhang anomalies. The dashed lines in the graphs plot the historical annual value-weighted alpha of each anomaly decile. The sample period is 1963 to 2015. The solid lines plot the alphas predicted by the model. (Color figure can be viewed at wileyonlinelibrary.com)

Prospect Theory: Revealed Preferences from Mutual Fund Flows

- Previous studies mainly test the asset pricing implications of prospect theory
- But prospect theory also makes predictions about investor demand for certain assets
- Han, Sui, and Yang (JFE 2026) evaluate prospect theory using mutual fund flows, which directly reveal investor preference/demand across assets
- They document that funds whose past returns generate higher prospect theory value attract significantly larger future flows

My Research: Stock Splits as Behavioral Signals

- **Li, Cui, Pang, and Xie (Financial Management, forthcoming)**
  - We try to explain why stock splits can be a credible positive signal to markets, even though the costs of implementing stock splits are trivial
  - Model: retail investors are loss averse and treat stock splits as good news
  - A split raises expectations about future firm fundamentals and price appreciation, while missing those expectations triggers disproportionately large price declines
  - In equilibrium, only managers with favorable information use splits to signal: loss aversion makes splits credible
  - Evidence from China: investors turn more optimistic after splits, higher split ratios bring stronger market reactions, and splitting firms' prices fall harder when they miss elevated expectations
- **Connection:** an application of loss aversion (this lecture) to corporate events: splits work as signals *because* investors evaluate outcomes as gains and losses relative to an elevated reference point

The Aggregate Stock Market

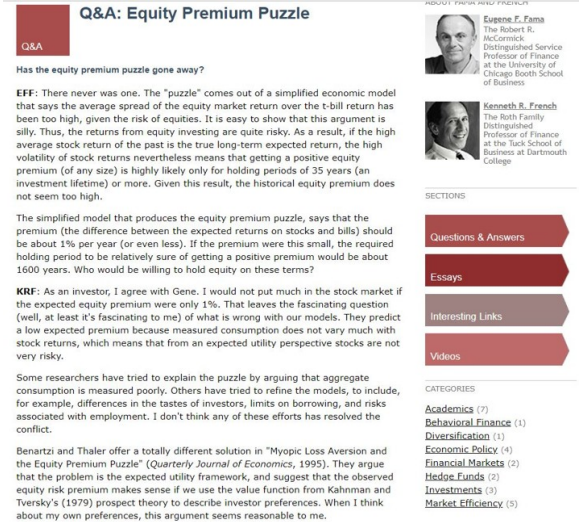
- Can prospect theory help us understand the properties of, and attitudes to, the aggregate stock market?
  - Benartzi and Thaler (1995) note that a model in which investors are loss averse over annual changes in the value of their stock market holdings predicts a large equity premium
  - Three elements:
    - Loss aversion
    - Annual evaluation of gain and loss (why?)
    - Narrow framing
  - Benartzi and Thaler (1995) emphasize the first two elements – "myopic loss aversion"

The Aggregate Stock Market, ctd.

Benartzi and Thaler (1995):

- The prospect theory value of the historical distribution of annual US stock market returns is approximately equal to the prospect theory value of the historical distribution of annual US bond market returns
- On the one hand, the greater volatility of stock market returns serves to lower its prospect theory value relative to that of bond market returns
- On the other hand, the much higher average return of the stock market serves to increase the prospect theory value of the stock market
- Benartzi and Thaler (1995) show that these two forces cancel out
- The stock market earns a high average return so that it can be competitive with the bond market in the eyes of prospect theory investors

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Q&A

Q&A: Equity Premium Puzzle

Has the equity premium puzzle gone away?

EFF: There never was one. The "puzzle" comes out of a simplified economic model that says the average spread of the equity market return over the t-bill return has been too high, given the risk of equities. It is easy to show that this argument is silly. Thus, the returns from equity investing are quite risky. As a result, if the high average stock return of the past is the true long-term expected return, the high volatility of stock returns nevertheless means that getting a positive equity premium (of any size) is highly likely only for holding periods of 35 years (an investment lifetime) or more. Given this result, the historical equity premium does not seem too high.

The simplified model that produces the equity premium puzzle, says that the premium (the difference between the expected returns on stocks and bills) should be about 1% per year (or even less). If the premium were this small, the required holding period to be relatively sure of getting a positive premium would be about 1600 years. Who would be willing to hold equity on these terms?

KRF: As an investor, I agree with Gene. I would not put much in the stock market if the expected equity premium were only 1%. That leaves the fascinating question (well, at least it's fascinating to me) of what is wrong with our models. They predict a low expected premium because measured consumption does not vary much with stock returns, which means that from an expected utility perspective stocks are not very risky.

Some researchers have tried to explain the puzzle by arguing that aggregate consumption is measured poorly. Others have tried to refine the models, to include, for example, differences in the tastes of investors, limits on borrowing, and risks associated with employment. I don't think any of these efforts has resolved the conflict.

Benartzi and Thaler offer a totally different solution in "Myopic Loss Aversion and the Equity Premium Puzzle" (*Quarterly Journal of Economics*, 1995). They argue that the problem is the expected utility framework, and suggest that the observed equity risk premium makes sense if we use the value function from Kahnman and Tversky's (1979) prospect theory to describe investor preferences. When I think about my own preferences, this argument seems reasonable to me.

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The role of narrow framing

- Barberis, Huang, and Santos (2001) find that narrow framing is a critical ingredient
  - Loss aversion over total wealth fluctuations does not produce a significant equity premium
  - Nor will it generate non-participation in the stock market
  - The stock market has low correlation with other risks

The role of probability weighting

- De Giorgi and Legg (2012) introduce probability weighting and show that it can significantly increase the equity premium
  - Because the aggregate stock market is negatively skewed: it is subject to occasional large crashes
  - Under probability weighting, investors overweight this left-tail outcome, which makes the stock market even less appealing

Ambiguity Aversion

- Economists have long distinguished between "risk" and "uncertainty"
  - Risk: the decision-maker is able to assign probabilities to the various possible outcomes
  - Uncertainty: not able to assign probabilities to these outcomes
- **The Ellsberg paradox, 1961**
- Two urns:
  - Urn C: 100 balls, 50 Red, 50 Black
  - Urn U: 100 balls, either Red or Black, unknown distribution
- Choose between:
  - Bet R1: draw a ball from urn C, get $100 if Red
  - Bet R2: draw a ball from urn U, get $100 if Red
- Choose between:
  - Bet B1: draw a ball from urn C, get $100 if Black
  - Bet B2: draw a ball from urn U, get $100 if Black
- The modal participant chooses both R1 and B1, suggesting that people act as if Urn C contains more red balls and more black balls than Urn U

Ambiguity Aversion: Applications

- Ambiguity aversion has been applied in finance in a number of ways (Epstein and Schneider, 2010).
- **Non-participation:**
  - The return on the stock market is more ambiguous than the return on a bank deposit or on Treasury Bills (Dow and Werlang, 1992)
- **Equity premium puzzle:**
  - If investors are ambiguity averse, they will require a much higher average return on the stock market than on Treasury Bills (Maenhout, 2004)
- **Under-diversification:**
  - Individuals view the returns of the domestic stock market, locally-headquartered stocks, and their own company stock as less ambiguous than foreign stock markets, distant stocks, and non-company stocks, respectively (Uppal and Wang, 2003)

Appendix: Prospect Theory and Stock Returns (Barberis, Mukherjee, Wang, RFS 2016) — Section 1: Conceptual Framework

Two steps: representation and valuation

Joint hypothesis

- Some investors mentally represent a stock by the **distribution of its past returns**.
- They evaluate this distribution of gains and losses according to **prospect theory** (Tversky–Kahneman 1992).

Valuation is straightforward: Tversky and Kahneman (1992) provide the formulas.

Representation is the hard step: how do investors represent a stock in their minds?

Our answer: the distribution of **monthly returns in excess of the market over the past five years**.

Main empirical prediction (tested in Section 2)

Stocks whose past return distributions have high (low) prospect theory values will have low (high) subsequent returns, on average — primarily among small-cap stocks.

1.1 Prospect Theory: Setup

- Cumulative prospect theory (Tversky–Kahneman 1992) resolves two limitations of the original 1979 version: applies to gambles with more than two nonzero outcomes; no dominated-gamble choices.
- Consider the gamble

$$(x_{-m}, p_{-m}; \ldots; x_{-1}, p_{-1}; x_0, p_0; x_1, p_1; \ldots; x_n, p_n), \tag{1}$$

read as "gain or lose x_{-m} with probability p_{-m}, x_{-m+1} with probability p_{m+1}, and so on," where $x_i < x_j$ for $i < j$, $x_0 = 0$ (so x_{-m}–x_{-1} are losses, x_1–x_n gains), and $\sum_{i=-m}^{n} p_i = 1$.

Example: a 50:50 bet to win \$110 or lose \$100 is $(-\$100, \frac{1}{2}; \$110, \frac{1}{2})$.

Expected utility individual with utility $U(\cdot)$ and wealth W computes

$$\sum_{i=-m}^{n} p_i \, U(W + x_i). \tag{2}$$

Cumulative prospect theory individual assigns the value

$$\sum_{i=-m}^{n} \pi_i \, v(x_i), \tag{3}$$

1.1 Prospect Theory: Decision Weights

$$\pi_i = \begin{cases} w^+(p_i + \ldots + p_n) - w^+(p_{i+1} + \ldots + p_n), & 0 \le i \le n, \\ w^-(p_{-m} + \ldots + p_i) - w^-(p_{-m} + \ldots + p_{i-1}), & -m \le i < 0, \end{cases} \tag{4}$$

where $v(\cdot)$ is the **value function**, $w^+(\cdot)$ and $w^-(\cdot)$ the **probability weighting functions**. (Boundary cases: $\pi_n = w^+(p_n)$, $\pi_{-m} = w^-(p_{-m})$.)

Intuition for $x_i \ge 0$: weight = weighted probability of all outcomes $\ge x_i$ minus that of outcomes $> x_i$. For losses: weighted probability of outcomes $\le x_i$ minus that of outcomes $< x_i$.

1.1 Functional Forms (Tversky–Kahneman 1992)

Value function:

$$v(x) = \begin{cases} x^{\alpha}, & x \geq 0, \\ -\lambda(-x)^{\alpha}, & x < 0, \end{cases} \tag{5}$$

Probability weighting functions:

$$w^{+}(P) = \frac{P^{\gamma}}{(P^{\gamma} + (1-P)^{\gamma})^{1/\gamma}}, \qquad w^{-}(P) = \frac{P^{\delta}}{(P^{\delta} + (1-P)^{\delta})^{1/\delta}}, \tag{6}$$

where $\alpha, \gamma, \delta \in (0,1)$ and $\lambda > 1$; and $v(0) = 0$, $w^{+}(0) = w^{-}(0) = 0$, $w^{+}(1) = w^{-}(1) = 1$.

Estimated parameters (median subject): $\alpha = 0.88$, $\lambda = 2.25$, $\gamma = 0.61$, $\delta = 0.69$.

$\Rightarrow$ To a first approximation, $v(\cdot)$ is piecewise-linear; concavity/convexity plays only a minor role.

1.1 Four Differences from Expected Utility

1. **Carriers of value**: gains and losses, not final wealth — argument of $v(\cdot)$ is x_i, not $W + x_i$.
2. **Loss aversion**: $v(\cdot)$ is kinked at the origin; more sensitive to losses than to gains of the same magnitude. Severity governed by λ (est. 2.25); captures aversion to $(-\$100, \frac{1}{2}; \$110, \frac{1}{2})$.
3. **Diminishing sensitivity**: $v(\cdot)$ concave over gains, convex over losses.
4. **Probability weighting**: objective probabilities replaced by transformed weights w^+, w^-; most extreme outcomes x_{-m}, x_n get weights $w^-(p_{-m})$, $w^+(p_n)$.

Main consequence: overweighting of tails

For $\gamma, \delta \in (0,1)$, $w^-(P) > P$ and $w^+(P) > P$ for low positive P. If p_{-m}, p_n small: $w^-(p_{-m}) > p_{-m}$, $w^+(p_n) > p_n$ — tail outcomes are overweighted. Captures simultaneous demand for lotteries *and* insurance: prefer $(\$5000, 0.001)$ to a certain \$5, but prefer a certain loss of \$5 to $(-\$5000, 0.001)$. Lower $\gamma, \delta \Rightarrow$ more overweighting.

1.2 Construction of Return Distributions

- Representation = **distribution of monthly returns over the past five years**, guided by how price charts appear in investment handbooks (e.g., Value Line): average window ≈ 5 years; monthly fluctuations are discernible, daily/weekly are not.
- Judgment & decision-making evidence: people passively accept the representation put in front of them.
- Returns: benchmark uses **excess returns over the market**; robustness with raw returns and excess of risk-free rate.
- Procedure: record each of the previous **60 monthly excess returns**, sort ascending: most negative r_{-m}, ..., most positive r_n, with $n = 60 - m$.

Historical return distribution (equal probability $\frac{1}{60}$ on each return):

$$\left(r_{-m}, \tfrac{1}{60};\ r_{-m+1}, \tfrac{1}{60};\ \ldots;\ r_{-1}, \tfrac{1}{60};\ r_1, \tfrac{1}{60};\ \ldots;\ r_n, \tfrac{1}{60}\right). \tag{7}$$

1.2 The TK Measure

Prospect theory value of the distribution in (7):

$$\mathrm{TK} \equiv \sum_{i=-m}^{-1} v(r_i)\left[w^-\left(\tfrac{i+m+1}{60}\right) - w^-\left(\tfrac{i+m}{60}\right)\right] + \sum_{i=1}^{n} v(r_i)\left[w^+\left(\tfrac{n-i+1}{60}\right) - w^+\left(\tfrac{n-i}{60}\right)\right]. \qquad (8)$$

- "TK" stands for **Tversky and Kahneman (1992)**.
- Parameters: $\alpha = 0.88$, $\lambda = 2.25$, $\gamma = 0.61$, $\delta = 0.69$; later estimates (Gonzalez–Wu 1999; Abdellaoui 2000) similar.
- TK does not depend on the *time order* of the 60 returns — a quick, passive reaction to the chart as an integral whole.

Time-discounted variant: if $t(i)$ = number of months ago that r_i was realized,

$$\mathrm{TK}(\rho) \equiv \frac{1}{\varphi}\sum_{i=-m}^{-1} \rho^{t(i)} v(r_i)\left[w^-\left(\tfrac{i+m+1}{60}\right) - w^-\left(\tfrac{i+m}{60}\right)\right] + \frac{1}{\varphi}\sum_{i=1}^{n} \rho^{t(i)} v(r_i)\left[w^+\left(\tfrac{n-i+1}{60}\right) - w^+\left(\tfrac{n-i}{60}\right)\right] \qquad (9)$$

where $\rho \in (0,1)$ and $\varphi = \rho + \ldots + \rho^{60}$.

1.3 Model: Setup and Traders

- Mean-variance framework. Risk-free asset with fixed return r_f; J risky assets, $j \in \{1, \ldots, J\}$; asset j has return $\tilde{r}_j$ with mean μ_j, s.d. σ_j; covariance $\sigma_{i,j}$. For portfolio p: $\tilde{r}_p$, μ_p, σ_p, $\sigma_{p,q}$.
- **Type 1:** traditional mean-variance investors holding the tangency portfolio ω_t (return $\tilde{r}_t$).
- **Type 2: prospect theory investors.** Start from ω_t and tilt toward high-TK, away from low-TK stocks:

$$\omega_p = \omega_t + k\,\omega_{\mathrm{TK}}, \qquad k > 0, \tag{10}$$

$$\omega_{\mathrm{TK},j} = \mathrm{TK}_j - \overline{\mathrm{TK}}, \qquad \overline{\mathrm{TK}} = \frac{1}{J}\sum_{j=1}^{J} \mathrm{TK}_j, \tag{11}$$

where TK_j is the prospect theory value (8) of stock j's 60 past monthly returns in excess of the market.

If the fraction of traditional investors is π (fraction $1-\pi$ of PT investors), the market portfolio is

$$\omega_m = \pi\omega_t + (1-\pi)\,(\omega_t + k\omega_{\mathrm{TK}}) = \omega_t + \eta\,\omega_{\mathrm{TK}}, \qquad \eta = (1-\pi)k. \tag{12}$$

1.3 Model: Pricing Proposition

Proposition 1

With β_x the market beta of asset/portfolio x, and "portfolio TK" the portfolio with weights ω_{TK}, the mean return μ_j of asset j satisfies

$$\frac{\mu_j - r_f}{\mu_m - r_f} = \beta_j - \frac{\eta\, s_{j,\text{TK}}}{\sigma_m^2\,(1 - \eta\,\beta_{\text{TK}})}, \tag{13}$$

where $s_{j,\text{TK}} = \text{Cov}(\tilde{\varepsilon}_j, \tilde{\varepsilon}_{\text{TK}})$ from the market-model regressions

$$\tilde{r}_j = r_f + \beta_j(\tilde{r}_m - r_f) + \tilde{\varepsilon}_j, \tag{14}$$

$$\tilde{r}_{\text{TK}} = r_f + \beta_{\text{TK}}(\tilde{r}_m - r_f) + \tilde{\varepsilon}_{\text{TK}}. \tag{15}$$

Under the additional assumption $\text{Cov}(\tilde{\varepsilon}_i, \tilde{\varepsilon}_j) = 0$ for $i \neq j$:

$$\frac{\mu_j - r_f}{\mu_m - r_f} = \beta_j - \frac{\eta\,\omega_{\text{TK},j}\,s_j^2}{\sigma_m^2\,(1 - \eta\,\beta_{\text{TK}})} = \beta_j - \frac{\eta\left(\text{TK}_j - \overline{\text{TK}}\right)s_j^2}{\sigma_m^2\,(1 - \eta\,\beta_{\text{TK}})}, \tag{16}$$

where $s_j^2 = \text{Var}(\tilde{\varepsilon}_j)$ in (14).

1.3 Model: Testable Prediction and Robustness

Prediction tested in Section 2

Eq. (16): stocks with higher TK_j than average have lower alphas — the prospect theory value of the past return distribution negatively predicts subsequent returns in the cross-section.

Intuition: high-TK stocks appeal to PT investors $\Rightarrow$ they tilt toward them $\Rightarrow$ overvaluation $\Rightarrow$ low subsequent returns.

Multiperiod extension: the expected *change* in a stock's TK over the next month should positively predict its return (a very good return 60 months ago rolls out of the window).

However, simulations show this second prediction is not robust to small misspecifications (e.g., assuming 4 instead of 5 years of data). The one-period prediction — TK negatively predicts subsequent returns — **is** robust. We therefore focus on the latter.

Summary of Barberis, Mukherjee, Wang (2016)

1. **1.1 Prospect theory:** value $\sum_i \pi_i v(x_i)$ with kinked value function (loss aversion, diminishing sensitivity) and probability weighting (tail overweighting); parameters $\alpha = 0.88$, $\lambda = 2.25$, $\gamma = 0.61$, $\delta = 0.69$.
2. **1.2 Construction:** mental representation = 60 monthly returns in excess of the market over the past five years; the TK measure in Eq. (8) computes the prospect theory value of this distribution.
3. **1.3 Model:** mean-variance economy with traditional and prospect-theory traders; Proposition 1 (Eqs. 13–16) implies high-TK stocks earn low subsequent returns.